

Shaharyar Shakir

DevOps Engineer | Cloud Infrastructure | CI/CD Automation | MLOps

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ACHIEVEMENTS

- Built and deployed a complete end-to-end DevOps pipeline using Jenkins, AWS, and Docker — reducing deployment time significantly.
- Hands-on experience with core AWS services: EC2, EKS, ECS, RDS, S3, and Route 53.
- Designed cloud infrastructure using Infrastructure as Code (IaC) with AWS CloudFormation.
- Practical experience with container orchestration using Docker and Kubernetes (K3s).
- Built a full MLOps pipeline for YouTube Sentiment Analysis — from automated ML training (DVC) to model serving (FastAPI) and a live Chrome Extension frontend.

EDUCATION

Bachelor of Science in Computer Science (Ongoing)

Expected Graduation: 2028

Relevant Coursework: Operating Systems, Computer Networks, Data Structures & Algorithms, Database Systems, Cloud Computing

PROJECT EXPERIENCE

YouTube Sentiment Analysis — MLOps Pipeline

2025

An end-to-end MLOps project that predicts sentiment of YouTube comments, featuring an automated ML pipeline, a FastAPI serving backend, and a live Chrome Extension frontend.

- Orchestrated an automated ML pipeline (data ingestion, preprocessing, training, evaluation) using DVC (Data Version Control) with reproducible, versioned stages defined in dvc.yaml.
- Tracked model experiments, parameters, and metrics using MLflow; registered the best-performing LightGBM model for production serving.
- Built a FastAPI backend serving the LightGBM + TF-IDF model with endpoints for sentiment prediction, pie chart generation, word cloud creation, and time-series trend graphs.
- Developed a Chrome Extension (HTML/CSS/JS) that scrapes comments from any YouTube video, calls the API, and displays interactive sentiment distributions and word clouds in-browser.
- Managed infrastructure with Docker Compose for the MLflow tracking server and Prometheus for API monitoring.
- Used uv for fast, reproducible Python dependency management across the entire project.

Tech Stack: Python, LightGBM, Scikit-Learn, NLTK, DVC, MLflow, FastAPI, Uvicorn, Docker, Prometheus, JavaScript

VProfile – End-to-End DevOps Project

2024

A production-grade multi-tier application deployment showcasing the full DevOps lifecycle.

- Designed and implemented a full CI/CD pipeline using Jenkins, automating build, test, and deployment stages.
- Deployed a Java web application on AWS EC2 using Tomcat as the application server and Nginx as a reverse proxy.
- Integrated RabbitMQ for message queuing, Memcached for caching, and AWS RDS for managed database services.
- Managed DNS routing and domain configuration using AWS Route 53 for reliable traffic distribution.
- Provisioned local development environments using Vagrant for consistent and reproducible dev/test workflows.

TECHNICAL SKILLS

Cloud Platforms	Amazon Web Services (EC2, EKS, ECS, RDS, S3, Route 53, CloudFormation)
CI/CD & MLOps	Jenkins, DVC, MLflow, Git, GitHub
Containerization	Docker, Docker Compose, Kubernetes (K3s)
Infrastructure as Code	AWS CloudFormation
Machine Learning	Scikit-Learn, LightGBM, NLTK, TF-IDF
Backend & APIs	FastAPI, Uvicorn, Prometheus
Web Servers	Nginx, Apache Tomcat
Operating Systems	Linux (Ubuntu, CentOS), Windows
Databases	MySQL, AWS RDS
Languages & Tools	Python, JavaScript, HTML/CSS, Bash, Vagrant, Postman, uv

CERTIFICATIONS & CONTINUOUS LEARNING

- Completed hands-on AWS project covering EC2, ECS, EKS, RDS, S3, Route 53, and CloudFormation.
- Actively pursuing AWS Cloud Practitioner / Solutions Architect certification.
- Self-directed learning in DevSecOps, monitoring (Prometheus/Grafana), and Terraform.

OBJECTIVE

Motivated Computer Science student and aspiring DevOps/MLOps Engineer with hands-on experience building CI/CD pipelines, deploying applications on AWS, managing containerized environments, and delivering end-to-end machine learning systems. Eager to contribute to a team that values automation, reliability, and cloud-native practices while continuing to grow across the full DevOps and MLOps toolchain.